Label Budget Strategies for Semi-Supervised Domain Adaptation: Selection and Allocation

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Abstract

TBC

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1. Introduction

1.1. Motivation

While Deep learning models for Image Classification and Semantic segmentation have played important roles in various application domains, their performance typically relies on the availability of a large volume of labelled data; In practice, deploying the desired models requires the collection of such labelled data, which can prove to be costly, time-consuming, and impractical. Consequently, practitioners often work with strict annotation budgets and must carefully consider the allocation of limited labelling resources.

This challenge is especially evident in domain adaptation settings, where a model trained on a labelled source domain is applied to a target domain with different data characteristics. Although the labelled samples may be available in the source domain, only a small amount of them are typically available in the target domain; As model performance is evaluated on the target domain, given the potentially significant impact a small number of target labels can have on results, the choice of which target samples to label is particularly important.

Despite this practical constraint, many existing approaches still implicitly label the target samples. Although the experimental evaluation can be simplified, the realistic deployment setting, where labelling decisions are deliberately made under resource constraints is not reflected. Consequently, the ability to effectively utilise limited target-domain labels continues to be a significant practical problem.

1.2. The Big Picture

Domain adaptation (DA) is a field that is commonly studied under different levels of target supervision, including unsupervised (UDA) and semi-supervised settings (SSDA). In this work, we focus on semi-supervised domain adaptation (SSDA) , where a model is trained using labelled source data, a small number of labelled target samples, and a larger number of unlabelled target data.

In SSDA, the labelled target subset can be obtained passively (given by the author or randomly labelled) or actively (queried under a budget). We adopt an SSDA training recipe and study the active construction of the labelled target set, which is commonly discussed under Active Domain Adaptation (ADA).

Recent SSDA methods have focused on utilising unlabelled target data through techniques such as pseudo-labelling or consistency regularisation, and have demonstrated strong performance on standard benchmarks. However, the majority of SSDA approaches treat the labelled target samples as a fixed subset obtained through random sampling. This assumption fails to address a critical aspect of real-world deployment: when annotation resources are limited, the selection of which target samples to label becomes a decision that can directly influence model performance. In addition, how a limited label budget is allocated across different classes may affect both overall accuracy and class-level behaviour.

From this perspective, our work remains within the SSDA learning setting, but adopts an active label acquisition viewpoint (ADA) , where the labelled target subset is constructed deliberately under a fixed budget rather than assumed to be given. As a result, analysing label selection and allocation explicitly allows us to better capture the impact of realistic labelling decisions in deployment scenarios.

1.3. Research question

The primary aim of this project is to investigate how different strategies for using a limited number of labelled target samples affect performance in semi-supervised domain adaptation (SSDA) for image classification (or semantic segmentation). Instead of proposing new model architectures, this work focuses on understanding how practical labelling decisions influence adaptation performance when annotation resources are limited.

Specifically, the project investigates two main aspects of target-domain labelling approaches. The first concerns sample selection deciding which specific target samples should be labelled under a fixed annotation budget; The second concerns label allocation, which is about how a limited number of labels should be distributed across different classes. The project's objective is to illuminate the distinct and collective influence of these factors on target-domain performance through a comprehensive examination of their individual and combined effects.

Based on these considerations, this project addresses the following research questions:

- Given a fixed total number of labelled target samples, how does target-domain performance vary across different target sample selection strategies?
- Under a fixed annotation budget, how does the allocation of labels across different classes influence overall target-domain accuracy and class-level performance?
- How does target-domain performance scale as the total target label budget increases (for example, 0.5%, 1%, 2%, and 5% of target samples labelled)?

Due to compute/time constraints, we will validate our label-budget strategies in a classification DA setting first (for rapid iteration), and extend to semantic segmentation if resources allow.

2. Related Work

2.1. Domain Adaptation, and UDA

Real-world deployment of computer vision models often involves a domain shift, whereby the distributions of the training and test data differ, resulting in significant reductions in performance. Domain Adaptation (DA) addresses this issue by transferring knowledge from a labelled source domain to a different target domain (Farahani et al., 2020). A common and challenging scenario is Unsupervised Domain Adaptation (UDA), where the target domain has no labels, prompting the development of methods that align representations across domains (Ahmed et al., 2023). Classic deep UDA approaches include adversarial feature alignment, as in Domain-Adversarial Neural Networks (DANN) (Ganin et al., 2016), and statistical alignment methods, such as Deep CORAL (Sun & Saenko, 2016). More recent work also considers settings where source data may be unavailable (source-free UDA), highlighting practical constraints and reliance on self-training and pseudo-labelling for adaptation (Fang et al., 2024).

2.2. Semi-Supervised Domain Adaptation (SSDA)

While UDA assumes no target labels, Semi-Supervised Domain Adaptation (SSDA) is often more practical. It uses a small number of labelled target samples together with many unlabelled target samples (and typically labelled source data). SSDA methods commonly combine DA-style alignment with semi-supervised learning techniques, such as pseudo-labelling, consistency regularisation and prototype-based alignment. For instance, ProML focuses on making more effective use of scarce labelled target data through prototype-based multi-level learning (Huang et al., 2023). Another approach reframes SSDA by adapting the source to better match the target distribution in order to reduce class mismatch issues during alignment (Yu & Lin, 2023).

However, the performance of SSDA depends strongly on how informative and representative the small labelled target subset is. If the labelled target samples lack class coverage or are biased towards 'easy' or overrepresented modes, the model may overfit that subset and fail to generalise across the target distribution. Furthermore, SSDA pipelines often rely on pseudo-labels, which can exacerbate errors when the model is miscalibrated due to domain shift, creating a harmful feedback loop. Recent SSDA work explicitly combines self-training, consistency and contrastive components to stabilise learning with few target labels, reflecting the fragility of adaptation under limited supervision (Morales-Brotons et al., 2024). These limitations emphasise the importance of treating the labelled target set as a resource that should be carefully constructed within budget constraints.

2.3. Active Learning (AL) and Active Domain Adaptation (ADA)

Active Learning (AL) is the study of how to select which unlabelled samples to annotate within a limited labelling budget, typically balancing uncertainty, diversity and representativeness (Tharwat & Schenck, 2023). Active Domain Adaptation (ADA) applies this concept to DA, querying a small set of target labels to improve adaptation under domain shift instead of labelling target data arbitrarily. Besides, ADA can also be viewed as SSDA where the labelled target subset is acquired actively via a query strategy rather than passively via random labelling. Recent ADA methods propose specialised acquisition strategies that go beyond standard uncertainty sampling. For example, they leverage local context in the queried data (Sun et al., 2023) or explicitly balance uncertainty and diversity under a fixed labelling budget (Tian et al., 2025). While these works are closely related, they mostly introduce new selectors. In contrast, our focus is on the policy-level question of how selection interacts with budget allocation.

2.4. Label budget strategies

In the context of budgeted target labelling, it is helpful to distinguish between selection (i.e. which specific target samples to label) and allocation (i.e. how to distribute the labelling budget across rounds, classes or difficulty levels). Even in standard AL, suitable querying strategies can depend on budget size and the acquisition protocol, suggesting that 'best' selection is not universal across budget regimes (Hacohen et al., 2022). In SSDA/ADA, domain shift intensifies this issue: the queried target subset can become biased, affecting pseudo-labelling and alignment behaviour. While some recent SSDA work explicitly integrates target labelling decisions with pseudo-labelling (e.g. EFTL) (He et al., 2024), there is less systematic evidence on how allocation choices (e.g. one-shot vs. multi-round; balanced vs. unconstrained) affect outcomes when using a fixed SSDA training recipe. This highlights the need for a controlled study of selection–allocation interactions to address this practical research gap. Therefore, our work focuses on label budget strategy in SSDA/ADA, clarifying and empirically evaluating how selection and allocation jointly influence adaptation performance rather than proposing a new acquisition function.

3. Dataset and Task

3.1. Task Definition

The task addressed by this project is the image classification (and potentially semantic segmentation) within a domain adaptation setting. The model is trained with synthetic data before being required to predict a single object class label from a predefined set of categories, given an input image from the real dataset. The classification task is consistent across both the source and target domains, while the visual appearance of images varies between domains. We will validate our label-budget strategies in a classification DA setting first (for rapid iteration), and extend to semantic segmentation if resources allow.

3.2. Dataset Description

Experiments are conducted using the **VisDA-2017** dataset, a widely used benchmark for evaluating visual domain adaptation. The dataset consists of images from multiple object categories and introduces a domain shift between synthetic and real-world data. The source domain comprises synthetic images rendered from 3D object models, while the target domain consists of real images collected from natural environments.

3.3. Domain Adaptation Setup

The current project adopts the semi-supervised domain adaptation (SSDA) setting. All images in the source domain are annotated with ground-truth class labels and are employed during the training process. Conversely, a limited number of images in the target domain are assigned ground-truth labels, with the remaining target-domain images being designated as unlabelled data.

The model is trained using labelled source data together with a small labelled subset and a larger unlabelled set from the target domain. Performance is evaluated exclusively on target-domain data, following standard practice in domain adaptation benchmarks.

3.4. Evaluation Metric

Model performance is evaluated using classification accuracy on the target-domain test set, following the standard evaluation protocol for the VisDA-2017 dataset. Further metrics may be applied with the progress of the ongoing project.

4. Methodology

4.1. Problem Setup

This project investigates semi-supervised domain adaptation (SSDA) in a synthetic-to-real setting for image classification. Let

$$\mathcal{D}_s = \{(x_s, y_s)\}$$

denote the source domain, consisting of fully labelled synthetic images, and

$$\mathcal{D}_t = \mathcal{D}_t^l \cup \mathcal{D}_t^u$$

denote the target domain, where $\mathcal{D}_t^l$ represents a small labelled subset of real images, approximately 1% of the target data, and $\mathcal{D}_t^u$ represents a larger set of unlabelled target images. We keep the SSDA training pipeline fixed, and vary how $\mathcal{D}_t^l$ is constructed: passive sampling vs active querying (ADA-style).

The objective is to learn a classifier that generalises well to the target domain using labelled source data, a limited number of labelled target samples, and abundant unlabelled target data. Performance is evaluated exclusively on the target-domain test set following standard domain adaptation practice.

4.2. Model Architecture

All experiments employ a consistent neural network architecture to ensure fair comparison across methods. The model consists of a ResNet-18 or ResNet-50 backbone pretrained on ImageNet and a task-specific fully connected classification head, adapted to the number of classes in the target domain. No architectural modifications are introduced across experiments; all performance differences arise solely from variations in label selection and allocation strategies.

4.3. Baseline Training Pipeline

4.3.1. Source-Only Training

As an initial baseline, the model is trained using only labelled source-domain data. The backbone and classification head are optimised using the standard cross-entropy loss:

$$\mathcal{L}_{\text{src}} = \mathbb{E}_{(x_s, y_s)\sim\mathcal{D}_s}\left[\text{CE}(f(x_s), y_s)\right],$$

where $f(\cdot)$ denotes the classifier. This baseline quantifies the synthetic-to-real domain gap when no target-domain supervision is utilised.

4.3.2. Semi-Supervised Adaptation Baseline

Following source-only training, the model is adapted to the target domain using a standard SSDA approach. This involves fine-tuning the model on the small labelled target subset $\mathcal{D}_t^l$, generating pseudo-labels for samples in the unlabelled target set $\mathcal{D}_t^u$, and applying consistency regularisation on unlabelled target data. The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{src}} + \mathcal{L}_t^l + \lambda \mathcal{L}_{\text{pl}},$$

where $\mathcal{L}_t^l$ is the supervised cross-entropy loss on labelled target samples, $\mathcal{L}_{\text{pl}}$ is the pseudo-label consistency loss applied to unlabelled target data, and λ is a scalar hyperparameter controlling the contribution of the pseudo-label loss. Only pseudo-labels whose predicted confidence exceeds a threshold τ are retained during training to reduce the impact of noisy predictions. This corresponds to a FixMatch-style pseudo-label consistency objective (Sohn et al., 2020).

4.4. Label Budget Strategies: Selection and Allocation

While conventional SSDA methods assume that the labelled target subset is obtained via random sampling, this project explicitly treats target label acquisition as a decision problem under a fixed annotation budget. This formulation reflects realistic deployment scenarios where labelling resources are limited and must be allocated strategically.

4.4.1. Label Budget Constraint

A fixed total number of target labels is assumed throughout training. Under this constraint, the goal is to maximise target-domain classification performance by selecting and allocating labels in an active manner. Experiments will be conducted under multiple label budgets, being a small proportion of the whole target dataset, including 0.5%, 1%, 2%, and 5% of target samples being labelled, to analyse performance scaling.

4.4.2. Target Sample Selection Strategies

To construct the labelled target subset $\mathcal{D}_t^l$, several selection strategies are evaluated. Random sampling selects target samples uniformly at random and serves as a baseline. Uncertainty-based sampling selects samples with the highest predictive uncertainty, quantified using entropy

$$H(p) = -\sum_c p_c \log p_c,$$

where p_c denotes the predicted class probability for class c. Diversity-based sampling clusters feature representations of target samples using k-means and selects samples closest to cluster centroids to encourage broad coverage of the target distribution and the features are extracted from the pretrained/fine-tuned backbone. A hybrid approach first filters samples based on uncertainty and subsequently applies diversity-based selection to the filtered set.

4.5. Label Allocation Policies

In addition to sample selection, the distribution of labels across classes is investigated. Uniform allocation distributes labels evenly across all classes, whereas difficulty-aware allocation assigns more labels to classes exhibiting higher predictive uncertainty or lower confidence. Class-wise allocation is implemented using the model's current pseudo-label predictions to approximate class membership before querying.

4.6. Training Procedure

The overall training procedure is as follows. The classifier is initially trained on labelled source-domain data, after which feature representations are extracted from unlabelled target images. A labelled target subset is then selected using one of the defined strategies. The model is fine-tuned using the labelled target samples, and pseudo-labels are generated for the remaining unlabelled samples. Training continues using both supervised and pseudo-label losses, and performance is evaluated on a held-out target-domain test set.

4.7. Evaluation Metrics

Model performance is primarily evaluated using target-domain classification accuracy. Secondary analyses include per-class accuracy to assess class-wise effects of label selection and allocation. Additionally, accuracy is analysed as a function of the target label budget to study performance scaling.

4.8. Experimental Protocol

To ensure fair and reproducible evaluation, all methods use identical network architectures, optimisers, and training schedules. Results are reported as the mean and standard deviation over multiple random seeds, and a fixed pseudo-label confidence threshold is used across all experiments.

4.9. Expected Outcomes

This methodology is expected to quantify the impact of informed label selection under limited supervision, identify selection strategies that maximise accuracy per labelled target sample, and analyse class-level benefits arising from different label allocation policies.

4.10. Implementation Choices and Contingencies

- Task choice: we prioritise segmentation; if compute/time is insufficient, we will fall back to classification while keeping the same selection/allocation study.
- Dataset choice: final dataset depends on task feasibility (e.g., GTA5 to Cityscapes for segmentation; Office-Home/VisDA for classification).
- Allocation implementation: class-wise allocation uses pseudo-label (or cluster) membership estimates for unlabelled target samples.

5. Experiments

6. Results and Evaluation

7. Conclusion

References

Ahmed, Sk Miraj, Raychaudhuri, Dripta S., Oymak, Samet, and Roy-Chowdhury, Amit K. Chapter 5 - source distribution weighted multisource domain adaptation without access to source data. In Govindaraju, Venu, Srinivasa Rao, Arni S.R., and Rao, C.R. (eds.), *Deep Learning*, volume 48 of *Handbook of Statistics*, pp. 81–105. Elsevier, 2023. doi: https://doi.org/10.1016/bs.host.2022.12.001. URL https://www.sciencedirect.com/science/article/pii/S0169716122000566.

Fang, Yuqi, Yap, Pew-Thian, Lin, Weili, Zhu, Hongtu, and Liu, Mingxia. Source-free unsupervised domain adaptation: A survey. *Neural Netw.*, 174(C), June 2024. ISSN 0893-6080. doi: 10.1016/j.neunet.2024.106230. URL https://doi.org/10.1016/j.neunet.2024.106230.

Farahani, Abolfazl, Voghoei, Sahar, Rasheed, Khaled, and Arabnia, Hamid R. A brief review of domain adaptation. *CoRR*, abs/2010.03978, 2020. URL https://arxiv.org/abs/2010.03978.

Ganin, Yaroslav, Ustinova, Evgeniya, Ajakan, Hana, Germain, Pascal, Larochelle, Hugo, Laviolette, François, Marchand, Mario, and Lempitsky, Victor. Domain-adversarial training of neural networks, 2016. URL https://arxiv.org/abs/1505.07818.

Hacohen, Guy, Dekel, Avihu, and Weinshall, Daphna. Active learning on a budget: Opposite strategies suit high and low budgets. In Chaudhuri, Kamalika, Jegelka, Stefanie, Song, Le, Szepesvari, Csaba, Niu, Gang, and Sabato, Sivan (eds.), *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 8175–8195. PMLR, 2022. URL https://proceedings.mlr.press/v162/hacohen22a.html.

He, Jiujun, Liu, Bin, and Yin, Guosheng. Enhancing semi-supervised domain adaptation via effective target labeling. AAAI'24/IAAI'24/EAAI'24. AAAI Press, 2024. ISBN 978-1-57735-887-9. doi: 10.1609/aaai.v38i11.29130. URL https://doi.org/10.1609/aaai.v38i11.29130.

Huang, Xinyang, Zhu, Chuang, and Chen, Wenkai. Semi-supervised domain adaptation via prototype-based multi-level learning. In Elkind, Edith (ed.), *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI-23*, pp. 884–892. International Joint Conferences on Artificial Intelligence Organization, 8 2023. doi: 10.24963/ijcai.2023/98. URL https://doi.org/10.24963/ijcai.2023/98. Main Track.

Morales-Brotons, Daniel, Chrysos, Grigorios, Tzoumas, Stratis, and Cevher, Volkan. The last mile to supervised performance: Semi-supervised domain adaptation for semantic segmentation, 2024. URL https://arxiv.org/abs/2411.18728.

Sohn, Kihyuk, Berthelot, David, Li, Chun-Liang, Zhang, Zizhao, Carlini, Nicholas, Cubuk, Ekin D., Kurakin, Alex, Zhang, Han, and Raffel, Colin. Fixmatch: Simplifying semi-supervised learning with consistency and confidence, 2020. URL https://arxiv.org/abs/2001.07685.

Sun, Baochen and Saenko, Kate. Deep coral: Correlation alignment for deep domain adaptation, 2016. URL https://arxiv.org/abs/1607.01719.

Sun, Tao, Lu, Cheng, and Ling, Haibin. Local context-aware active domain adaptation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 18634–18643, 2023. Code: https://github.com/tsun/LADA.

Tharwat, Alaa and Schenck, Wolfram. A survey on active learning: State-of-the-art, practical challenges and research directions. *Mathematics*, 11(4):820, 2023. doi: 10.3390/math11040820. URL https://doi.org/10.3390/math11040820.

Tian, Qing, Li, Yanzhi, Yu, Jiangsen, Shen, Junyu, and Ou, Weihua. Rethinking active domain adaptation: Balancing uncertainty and diversity. *Image and Vision Computing*, 158:105492, 2025. ISSN 0262-8856. doi: https://doi.org/10.1016/j.imavis.2025.105492. URL https://www.sciencedirect.com/science/article/pii/S0262885625000800.

Yu, Yu-Chu and Lin, Hsuan-Tien. Semi-supervised domain adaptation with source label adaptation, 2023. URL https://arxiv.org/abs/2302.02335.